

Development and Validation of a Risk Calculator for Prediction of Cardiac Risk After Surgery

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Background—Perioperative myocardial infarction or cardiac arrest is associated with significant morbidity and mortality. The Revised Cardiac Risk Index is currently the most commonly used cardiac risk stratification tool; however, it has several limitations, one of which is its relatively low discriminative ability. The objective of the present study was to develop and validate a predictive cardiac risk calculator.

Methods and Results—Patients who underwent surgery were identified from the American College of Surgeons' 2007 National Surgical Quality Improvement Program database, a multicenter (>250 hospitals) prospective database. Of the 211 410 patients, 1371 (0.65%) developed perioperative myocardial infarction or cardiac arrest. On multivariate logistic regression analysis, 5 predictors of perioperative myocardial infarction or cardiac arrest were identified: type of surgery, dependent functional status, abnormal creatinine, American Society of Anesthesiologists' class, and increasing age. The risk model based on the 2007 data set was subsequently validated on the 2008 data set (n=257 385). The model performance was very similar between the 2007 and 2008 data sets, with C statistics (also known as area under the receiver operating characteristic curve) of 0.884 and 0.874, respectively. Application of the Revised Cardiac Risk Index to the 2008 National Surgical Quality Improvement Program data set yielded a relatively lower C statistic (0.747). The risk model was used to develop an interactive risk calculator.

Conclusions—The cardiac risk calculator provides a risk estimate of perioperative myocardial infarction or cardiac arrest and is anticipated to simplify the informed consent process. Its predictive performance surpasses that of the Revised Cardiac Risk Index. (*Circulation*. 2011;124:381-387.)

Key Words: myocardial infarction ■ cardiac arrest ■ risk ■ assessment ■ perioperative period

The incidence of perioperative myocardial infarction (MI) varies widely, ranging from 3% to 17%, with the incidence being higher in those with coronary artery disease (CAD) or with risk factors for CAD than in those without these risk factors.¹⁻³ It is expected that incidence will increase in the years to come, given that there will be a simultaneous increase in the incidence of risk factors for CAD with the aging population and that this is the population in which a majority of surgeries are performed.^{4,5} Perioperative cardiac events are a leading cause of death after surgery.⁵⁻⁷ Thus, risk stratification for perioperative MI and cardiac arrest (CA) has become an integral part of the workup of any patient before surgery.² It serves to identify patients who are at higher risk for perioperative MI or CA and who would benefit the most from cardiac optimization before elective surgery.^{2,6,8}

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Since Goldman et al first proposed a risk index in 1977, several perioperative cardiac risk indices/scores have been proposed.^{1,8-10} The most widely used of these risk indices is the Revised Cardiac Risk Index (RCRI); however, it has several limitations.⁸

With more than 200 hospitals reporting preoperative, intraoperative, and postoperative data to the American College of Surgeons' National Surgical Quality Improvement Program (NSQIP), NSQIP is the largest and most complete prospective, clinical, national surgical database.^{11,12} Therefore, an analysis of the NSQIP database was undertaken to ascertain risk factors associated with intraoperative/postoperative MI or CA (MICA) and to develop and validate a risk calculator for its prediction. Its

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performance was subsequently compared with that of the RCRI. None of the existing risk indices has a high predictive ability for MICA in patients undergoing aortic or noncardiac vascular surgery. Thus, even though the primary objective was to develop and validate a risk calculator to predict risk of MICA after all surgeries, the performance of the validated MICA risk calculator was further specifically tested among patients undergoing aortic or noncardiac vascular surgery to ascertain its predictive ability in that population.

Methods

Data Set

Data were extracted from the 2007 and 2008 NSQIP Participant Use Data Files.¹³ These are multicenter, prospective databases with 183 (year 2007) and 211 (year 2008) participating academic and community US hospitals and include data collected on 136 perioperative variables. The 2006 data set was not used because of slight differences in definitions. In NSQIP, a participating hospital's surgical clinical nurse reviewer captures data using a variety of methods, one of which is medical chart abstraction. The data are collected on the basis of strict criteria formulated by a committee. To ensure the data collected are of a high quality, the NSQIP has developed different training mechanisms for the surgical clinical nurse reviewer and conducts an interrater reliability audit of participating sites.¹³ The processes of surgical clinical nurse reviewer training, interrater reliability auditing, data collection, and sampling methodology have been described in detail.^{13–15} The combined results of the audits revealed an overall disagreement rate of approximately 1.99% for all assessed program variables. The Participant Use Data Files are Health Insurance Portability and Accountability Act–compliant data files that contain patient-level aggregate data and do not identify hospitals, healthcare providers, or patients. Trauma patients, transplant patients, and those younger than 16 years of age are excluded from the NSQIP database. The NSQIP database captures outcomes through 30 days after surgery, except for hospital length of stay, which is recorded until the patient is discharged.

Patients

Patients who underwent the surgeries listed in Table 1 were studied. Data on demographics, lifestyle, comorbidity, and other variables were obtained. Demographic variables analyzed included age, sex, and race. Lifestyle variables included smoking (within 1 year of surgery) and alcohol (more than 2 drinks per day in the 2 weeks before surgery). Comorbidities studied included presence or absence of renal disease (acute renal failure and dialysis dependence), CAD (angina within 30 days of surgery, MI within 6 months of surgery, prior percutaneous coronary intervention, and cardiac surgery), congestive heart failure, hypertension, peripheral vascular disease (revascularization/amputation for peripheral vascular disease and rest pain in lower extremity), pulmonary disease (ventilator dependence before surgery, history of chronic obstructive pulmonary disease, and preexisting pneumonia), sepsis (systemic inflammatory response syndrome, sepsis, and septic shock), liver disease (ascites or esophageal varices), neurological event or disease (stroke with or without residual deficit, transient ischemic attack, hemiplegia, paraplegia, quadriplegia, impaired sensation/delirium, coma, and central nervous system tumor), diabetes mellitus, disseminated cancer, chronic corticosteroid use, weight loss (>10% in the 6 months before surgery), bleeding disorders, and open wound. Other factors considered were American Society of Anesthesiologists' (ASA) class,¹⁶ preoperative functional status/ability to perform activities of daily living in the 30 days before surgery (independent, partially dependent, totally dependent), dyspnea (none, moderate exertion, at rest), body mass index, pregnancy, prior operation within 30 days, preoperative transfusion of >4 U of packed red blood cells, neoadjuvant chemoradiation, and emergency surgery. The complete definitions for all the above variables have been published previously.¹³

Preoperative laboratory variables analyzed included blood urea nitrogen, creatinine, albumin, bilirubin, hematocrit, platelet count, white blood cell count, partial thromboplastin time, and prothrombin

Table 1. Description of Surgeries

Type of Surgery	Organs/Systems Included	n (% of 211 410)
Anorectal	Anus and rectum with transanal approach	3265 (1.5)
Aortic	Aorta	4479 (2.1)
Bariatric	Bariatric surgeries	12 337 (5.8)
Brain	Brain	370 (0.2)
Breast	Breast	21 359 (10.1)
Cardiac	Heart	631 (0.3)
ENT	ENT and head and neck surgeries except thyroid and parathyroid	646 (0.3)
FG/HPB	Foregut and hepatopancreatobiliary surgeries: esophagus, stomach, duodenum, pancreas, liver, and biliary tree (except isolated cholecystectomy)	12 254 (5.8)
GBAAS	Gall bladder, appendix, adrenals or spleen; biliary tree surgeries other than cholecystectomy not included	35 940 (17.0)
Gyn	Obstetric and gynecological surgeries	2861 (1.4)
Hernia	Ventral, inguinal, femoral, and other hernias (not paraesophageal)	31 692 (15.0)
Intestine	Intestines below the level of duodenum	31 492 (14.9)
Neck	Thyroid and parathyroid surgeries	10 179 (4.8)
Nonesophageal thoracic	Thoracic surgeries excluding heart and esophagus	1324 (0.6)
Orthopedic	Orthopedics and nonvascular extremity	9272 (4.4)
Other abdomen	Abdominal surgeries not covered by above	4086 (1.9)
Peripheral vascular	Nonaortic, nonvein vascular surgeries	17 490 (8.3)
Skin	Skin	4906 (2.3)
Spinal	Spine	1469 (0.7)
Urology	Kidneys and urinary system	1975 (0.9)
Vein	Veins	3383 (1.6)

ENT indicates ear, nose, and throat; FG/HPB, foregut and hepatopancreatobiliary; GBAAS, gall bladder, appendix, adrenals, or spleen; and Gyn, gynecological.

time. NSQIP definitions of “normal” and “abnormal” were used to categorize laboratory values as normal and abnormal; missing data constituted a third categorical level, an indicator variable.¹⁷

Outcome

The primary end point was intraoperative/postoperative MI or CA through 30 days after surgery.

Cardiac arrest has been defined by the NSQIP as “The absence of cardiac rhythm or presence of chaotic cardiac rhythm that results in loss of consciousness requiring the initiation of any component of basic and/or advanced cardiac life support.” Patients with automatic implantable cardioverter defibrillator (AICD) that fire but do not lose consciousness are not included in this definition.¹⁸ Thus, it effectively includes all forms of malignant ventricular and supraventricular arrhythmias, along with pulseless electric activity and asystole.

MI has been defined in the NSQIP database as “Presence of one of the following: (1) documentation of electrocardiogram (ECG) changes indicative of acute MI (one or more of the following): (a) ST

elevation >1 mm in two or more contiguous leads (b) new left bundle branch (c) new Q-wave in two or more contiguous leads (2) new elevation in troponin greater than 3 times upper level of the reference range in the setting of suspected myocardial ischemia.”¹⁸

The secondary end point was intraoperative/postoperative MICA among patients undergoing aortic or noncardiac vascular surgery.

Statistical Analysis

Univariate exploratory analysis was performed on the 2007 NSQIP data set. Pearson χ^2 test or Fisher exact test were performed for the categorical variables, with t or F test for continuous variables. Stepwise multivariate logistic regression was performed to assess the most parsimonious combination of risk factors predictive of MICA, thus creating the full model. To address the possibility that the surgery type might interact with any of the other variables, we included all second-order interaction terms that involved surgery in the list of candidate variables.

To reduce the number of risk factors, we then created parsimonious models by sequentially removing variables from the full model. A forward-selection procedure was applied to select a predetermined number of variables for inclusion in the final logistic model from a list of candidate variables, which in this analysis included all preoperative variables. The number of variables to be included in the final model was determined as 5 because we found that the inclusion of more than 5 variables in the model did not significantly improve the prediction (C statistic) of MICA, but made the model more complex. Statistical analysis was performed with SAS (version 9.2; SAS Institute, Cary, NC). $P < 0.05$ was considered significant.

Risk Model Performance

The accuracy of a logistic regression model is usually assessed by its discrimination and calibration.¹⁹ Discrimination measures how well a model can distinguish between cases (who develop MICA) and noncases (who do not develop MICA). Discrimination is usually assessed by the C statistic, also known as the area under the receiver operating characteristic curve. This statistic evaluates each pair of observations that have different outcomes and calculates the proportion of times when the patient who developed MICA had a higher predicted risk of MICA than the patient who did not develop MICA. The C statistic ranges from 0.50 (no better than flipping a coin) to 1.00 (model is 100% correct).

Calibration measures the ability of a model to generate predictions that are on average close to the average observed outcome. The most widely used method for doing this in hospital mortality models is the Hosmer-Lemeshow test, which examines how well the percentage of observed MICA matches the percentage of predicted MICA over deciles of predicted risk.¹⁹ In studies with large sample sizes, it is suggested that a calibration graph of observed versus predicted events be constructed.¹⁹ If the model calibrates well, there will not be a substantial deviation from the 45° line of perfect fit.

Risk Model Validation

Once a model was trained on the basis of the 2007 data set, an independent data set (2008 data set) was used to validate the model. The model validation applied the trained model from the 2007 data set to estimate MICA probabilities for all patients in the 2008 data set. These estimated probabilities were then compared with actual MICA status in the 2008 data set by computing a C statistic (discrimination). To do this, a receiver operating characteristic curve was constructed on the basis of the sensitivity and specificity of the predictions from the 2007 model on the 2008 data set for various prediction cut points. The C statistic is equivalent to the area under this receiver operating characteristic curve and was computed by use of the trapezoidal rule. This C statistic reflects how much predictive accuracy the trained (2007) model has on the 2008 data set. If this C statistic shows favorable predictive accuracy, then the model is considered validated. As previously described in literature, similar results for calibration and discrimination indicate validation in an independent data set.^{20–23}

Table 2. Estimates, Standard Errors, and Variables Associated With Myocardial Infarction or Cardiac Arrest in Stepwise Logistic Regression Analysis (2007 NSQIP Data Set—Final Model)

Parameter	Estimate	SE	Adjusted OR	95% Wald CI
Intercept	−5.25	0.24		
Totally dependent functional status*	1.03	0.09	2.79	2.36–3.30
Partially dependent functional status*	0.65	0.08	1.92	1.65–2.23
ASA class 1†	−5.17	0.72	0.01	0.001–0.02
ASA class 2†	−3.29	0.17	0.04	0.03–0.05
ASA class 3†	−1.92	0.13	0.15	0.11–0.19
ASA class 4†	−0.95	0.12	0.39	0.30–0.49
Creatinine (abnormal)‡	0.61	0.06	1.84	1.63–2.09
Creatinine (missing)‡	−0.10	0.15	0.91	0.68–1.21
Age per year of increase	0.02	0.002	1.02	1.01–1.02
Anorectal§	−0.16	0.52	0.85	0.31–2.37
Aortic§	1.60	0.17	4.96	3.55–6.93
Bariatric§	−0.25	0.30	0.78	0.43–1.40
Brain§	1.40	0.42	4.04	1.79–9.13
Breast§	−1.61	0.47	0.20	0.08–0.50
Cardiac§	1.01	0.30	2.74	1.51–4.99
ENT§	0.71	0.73	2.04	0.49–8.47
Foregut/hepato-pancreatobiliary§	1.39	0.17	4.02	2.89–5.60
GBAAS§	0.59	0.18	1.81	1.27–2.58
Intestinal§	1.14	0.16	3.12	2.29–4.24
Neck§	0.18	0.29	1.20	0.68–2.12
Obstetric/gynecologic§	0.76	0.43	2.14	0.91–5.05
Orthopedic§	0.80	0.18	2.22	1.55–3.17
Other abdomen§	1.13	0.19	3.11	2.13–4.54
Peripheral vascular§	0.86	0.16	2.36	1.72–3.25
Skin§	0.54	0.25	1.72	1.06–2.79
Spine§	0.21	0.60	1.24	0.38–4.00
Thoracic§	0.40	0.42	1.49	0.67–3.32
Vein§	−1.09	1.01	0.34	0.05–2.43
Urology§	−0.26	0.52	0.77	0.28–2.14

NSQIP indicates National Surgical Quality Improvement Program; SE, standard error; OR, odds ratio; CI, confidence interval; ASA, American Society of Anesthesiologists; ENT, ear, nose, and throat; GBAAS, gallbladder, adrenal, appendix, spleen surgery.

Abnormal creatinine was creatinine >1.5 mg/dL.

The estimate and the SE refer to the estimate of the logistic regression coefficient for the specific variable and its associated SE. C statistic=0.884.

Reference groups were as follows: *Independent functional status; †ASA class 5; ‡normal creatinine; §hernia surgery. 0.4% of ASA class data were missing.

Development of Risk Calculator

Once the model was validated, it was used to develop the risk calculator, which took the form of an interactive spreadsheet that accepts patient covariate information and returns an estimated probability percentage of MICA based on the validated model. Alternatively, one can generate this estimated probability percentage using the model fit directly. The parameter estimates and standard errors for the model are presented in Table 2. These estimated coefficients can be used to estimate the logit ($\hat{L}$) for a patient using the standard binary logistic regression equation. The estimated

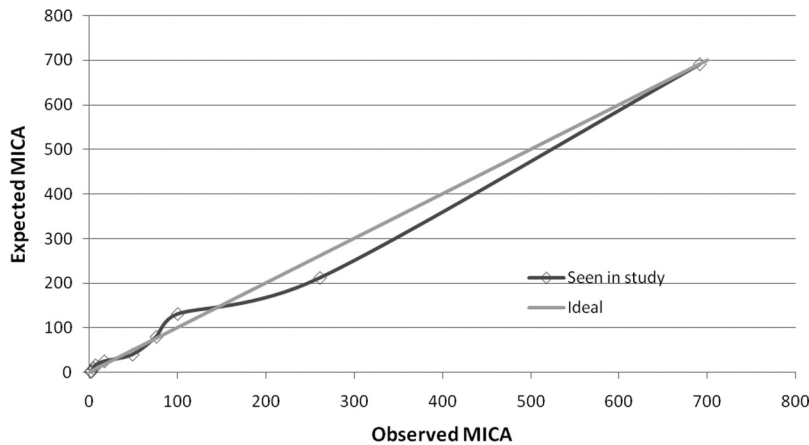


Figure 1. Observed vs expected perioperative myocardial infarction or cardiac arrest (MICA): calibration of predictions in the training set.

MICA probability percentage for a patient is then computed with the following formula:

$$\text{estimated probability} = 100\% \times \frac{e^L}{1 + e^L}$$

Comparison With Revised Cardiac Risk Index

To compare the risk calculator with the RCRI, the risk predictors in the RCRI were applied to the 2008 NSQIP data set with end points of MICA among all surgeries and MICA among vascular surgeries. Multivariate logistic regression analyses were performed. The C statistic (discrimination) was used to compare performance of the risk calculator with that of RCRI.

Results

Demographics

In the 2007 NSQIP data set ($n=211\,410$), MICA was seen in 1371 patients (0.65%) (intraoperative MI, 154 patients; postoperative MI, 357 patients; intraoperative CA, 24 patients; postoperative CA, 902 patients). Patients with MICA were significantly older than those without MICA (median 71 versus 56 years, $P<0.0001$). Men (57.5%) constituted a significantly higher proportion ($P<0.0001$) of those with MICA than women (42.7%; online-only Data Supplement Table I). In the 2008 data set, which was used for validation ($n=257\,385$), MICA was seen in 1401 patients (0.54%).

Univariate Analysis (2007 Data Set)

MICA was significantly associated with a multitude of variables ($P<0.0001$ for most; online-only Data Supplement Table I). Online-only Data Supplement Table II displays the intraoperative and postoperative differences in patients with and without MICA. Postoperatively, patients with MICA had more complications than those without MICA. Operative time and hospital length of stay were significantly longer ($P<0.0001$ for both). Death within 30 days was also significantly higher in patients with MICA (61.42% versus 1.35%, $P<0.0001$). Median number of days from surgery until MI was 2 (interquartile range 1 to 6 days), whereas the median number of days from surgery until CA was 4 (interquartile range 1 to 10 days).

Multivariate Analysis for MICA (2007 Data Set)

Preoperative variables significantly associated with an increased risk for MICA in the full model included ASA class, preoperative dialysis dependence, emergency case, male sex, dependent functional status, history of recent MI, black race, rest pain in

lower extremity due to peripheral vascular disease, $>10\%$ preoperative weight loss, history of transient ischemic attack, peripheral vascular disease requiring revascularization or amputation, prior percutaneous coronary intervention, hypertension, chronic obstructive pulmonary disease, dyspnea at rest or exertion, preoperative sepsis, increasing age, abnormal creatinine, blood urea nitrogen, hematocrit, and type of surgery. Preoperative variables significantly associated with an increased risk for MICA in the final model included ASA class, dependent functional status, increasing age, abnormal creatinine (>1.5 mg/dL), and type of surgery (Table 2). None of the second-order interaction terms involving surgery were chosen, which suggests that there was not a substantial interaction between surgery type and any of the other variables in the model. The variables in the model remained statistically significant even after Bonferroni correction.

Development and Validation of Risk Model

The 2007 data set was used as the training set to develop the model, and the 2008 data set served as the validation set. The risk model included significant predictors from the 2007 data set. The parameter estimates and their standard errors are summarized in Table 2.

Reference group for type of surgery was hernia surgery; for laboratory values, normal values; for functional status, independent functional status; and for ASA class, class 5. Table 2 can be used to generate probability estimates identical to the risk calculator by inserting the appropriate coefficient estimates into the standard logistic regression model to compute the estimated logit and then translating this logit into the probability scale as described in Methods.

The C statistic for the training set was 0.892 in the full model and 0.884 in the final model, which indicates excellent discrimination. Figure 1 shows that the calibration (Hosmer-Lemeshow goodness-of-fit test) was excellent, without a substantial deviation from the 45° line of perfect fit.

The selected risk model (final model) was then applied to the 2008 validation set. The C statistic that arose from use of the 2007 model to estimate MICA probability in the 2008 data set was 0.874, which indicates excellent discrimination. These findings indicate that the model performance was very similar in both the 2007 training set and the 2008 validation set, with the model continuing to have excellent discrimination in an inde-

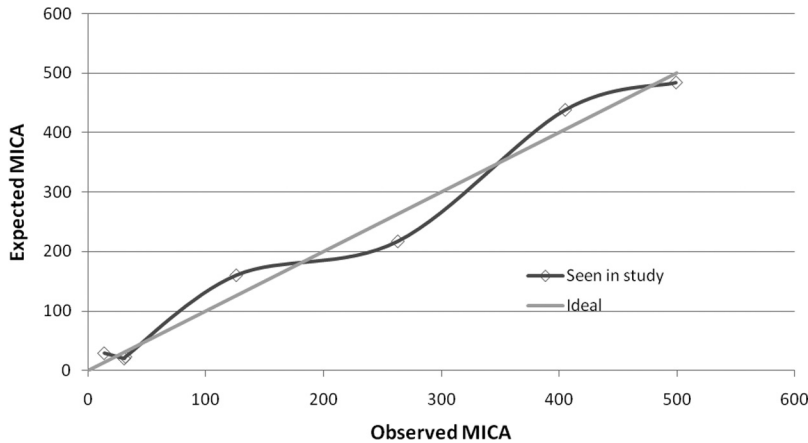


Figure 2. Observed vs expected perioperative myocardial infarction or cardiac arrest (MICA) for the Revised Cardiac Risk Index: calibration of predictions.

pendent data set. This method of validation using similar C statistics has been described previously in the literature.^{20–22}

Development of Risk Calculator

The selected model was then used to develop an interactive risk calculator. In the risk calculator, values are entered as 0 and 1 for absence or presence, respectively, of significant risk factors. In the case of continuous variables, values are entered as a number. When the required input is entered into this calculator for a given patient, it returns a model-based percent estimate of postoperative MICA.

We present a few examples of calculated postoperative MICA using the risk calculator:

1. Sixty-year-old man, ASA class 4, independent, with normal creatinine, undergoing aortic surgery: 3.18%.
2. Seventy-year-old woman, ASA class 3, independent, with normal creatinine, undergoing brain surgery: 1.22%.
3. Seventy-year-old man, ASA class 5, totally dependent, with elevated creatinine, undergoing pancreatoduodenectomy: 30.23%.
4. Forty-year-old woman, ASA class 2, independent, with normal creatinine, undergoing laparoscopic cholecystectomy: 0.08%.
5. Sixty-five-year-old man, ASA class 4, partially dependent, with elevated creatinine, undergoing peripheral vascular surgery: 5.75%.

Validation of Risk Model on Patients Undergoing Vascular Surgery

The risk model was further applied specifically to patients in the 2008 NSQIP data set who underwent aortic or other noncardiac vascular surgery ($n=26\,183$) to ascertain the discriminative ability of the model for MICA in patients undergoing these surgeries. The C statistic of the model was 0.746.

Application of Revised Cardiac Risk Index to National Surgical Quality Improvement Program Data Set

The RCRI identified 6 independent predictors of perioperative cardiac complications.⁸ When it was applied to the 2008 NSQIP data set with an end point of MICA among all surgeries, all of the 6 risk factors were significant: high-risk surgery odds ratio (OR) 2.01, 95% confidence interval (CI) 1.81 to 2.23; history of congestive heart failure OR 3.26, 95% CI 2.67 to 3.98; history of ischemic heart disease OR 3.02, 95% CI 2.51 to 3.64; history of stroke OR 1.92, 95% CI 1.67

to 2.20; preoperative treatment with insulin OR 1.27, 95% CI 1.10 to 1.46; and preoperative serum creatinine >2.0 mg/dL OR 4.86, 95% CI 4.31 to 5.49. The C statistic was 0.747. Figure 2 shows that the calibration (Hosmer-Lemeshow goodness-of-fit test) was moderate. Calibration for RCRI had not been mentioned by Lee et al originally.⁸ When the RCRI was applied to the 2008 NSQIP data set to assess risk of MICA in patients who underwent aortic or other noncardiac vascular surgery ($n=26\,183$), the C statistic was 0.591.

Discussion

In the present study, ASA class, dependent functional status, age, abnormal creatinine (>1.5 mg/dL), and type of surgery were associated with cardiac risk after surgery. Some of these are also components of existing risk indices.^{1,2,8,10,24} Dependent functional status was not a component of previous indices; however, a trend toward higher complication rates in patients with worse functional capacity was found in RCRI. Insulin-dependent diabetes mellitus was not a significant predictor on multivariate analysis in the present study; in the original study by Lee et al on RCRI,⁸ it was significant only for the derivation set and not the validation set. Surprisingly, a history of congestive heart failure was also not significantly associated with postoperative MICA in the present study. It is likely that heart failure is not predictive of MICA independent of functional status or ASA class.

Although many previous risk stratification tools exist, they have several limitations. RCRI, the most widely used risk index, was derived and validated on a relatively small cohort of 4315 patients in contrast to more than 400 000 patients in the present study.⁸ Furthermore, categorization of surgeries as used in the RCRI is not very relevant today because of advances in the field of surgery, especially minimally invasive surgery such as laparoscopy. For instance, in the RCRI, all suprainguinal vascular, intraperitoneal, and intrathoracic surgeries were classified as high risk, thus equating laparoscopic appendectomy and cholecystectomy to pancreatotomy and liver resections. This is not the case, as is evidenced by the higher risk of MICA in patients who underwent foregut/hepatopancreatobiliary (pancreatotomy/liver resection) surgery compared with patients who underwent laparoscopic appendectomy and cholecystectomy in the NSQIP data set. In the RCRI, each organ system was not considered individually. Evidently, the surgery-specific risk is independent of other factors, and hence, organ-based classification of

surgeries is the most appropriate approach for MICA risk assessment, providing a more precise estimate of risk.

Despite its relative ease of use, the RCRI lacks in its discriminative/predictive ability. It has been widely suggested that the discriminative ability of a risk index should be greater than 0.8 for a prediction model to be considered clinically relevant.^{25,26} The receiver operating characteristic area under the curve for RCRI, which represents discriminative ability, was 0.76 in the original study by Lee et al⁸ and 0.75 in a review by Ford et al,⁶ which makes its discriminative ability only moderate. RCRI was applied to the 2008 NSQIP data set to predict MICA among all surgeries, to assess its performance on an independent prospective cohort of patients across the country, and to then compare it to the risk calculator in the present study. Its C statistic was 0.75, lower than the 0.88 seen for the MICA risk calculator proposed in the present study. RCRI was also compared to the MICA risk calculator to assess discrimination for MICA specifically in patients who underwent aortic or other noncardiac vascular surgery. The C statistic for RCRI was again lower (0.59 versus 0.75).

Discrimination for the MICA risk calculator for aortic or other noncardiac vascular surgery is also higher than that of the Vascular Study Group of New England Cardiac Risk Index (0.75 versus 0.71).²⁷ The Vascular Study Group of New England Cardiac Risk Index had age, smoking, insulin-dependent diabetes mellitus, CAD, abnormal cardiac stress test, long-term β -blocker therapy, chronic obstructive pulmonary disease, and abnormal creatinine as significant risk factors. Of these, smoking and insulin-dependent diabetes mellitus were not significant in the MICA risk calculator, while stress test and β -blocker therapy were not a part of the NSQIP data set. CAD and chronic obstructive pulmonary disease, although a part of the full model, were not a part of the final model.

The MICA risk calculator was developed to aid in the surgical decision-making and informed consent process. It is in the form of an interactive spreadsheet and is available online at <http://www.surgicalriskcalculator.com/miorcardiacarrest> for free download. When the required input is entered into this calculator for a given patient, it returns a model-based percent estimate of MICA. In the risk calculator, values are entered as 0 and 1 for absence or presence, respectively, of the significant predictive factors or as the actual value for continuous variables.

Previous cardiac indices like the one proposed by Goldman et al¹ and the RCRI⁸ used logistic regression models to create point-based risk score systems. We instead chose to develop the risk calculator directly on the basis of the logistic regression model. This approach allowed direct modeling and prediction of MICA, rather than the use of 1 model to assess MICA and another to predict risk based on a point system. The predicted probabilities are based on the exact model used to assess MICA, rather than a model of a model, which is needed for a point system in this context. Hence, no loss of accuracy associated with the use of a second model is incurred with this strategy. In addition, the previous indices did not provide an actuarial estimate of risk; instead, patients were classified as being at high risk/intermediate risk/low risk for perioperative cardiac complications. With the MICA risk calculator, we instead provide an exact model-based estimate of MICA probability for a patient. This approach is more

precise than a point system, but it may be less simple for some users to implement. However, as clinicians take advantage of new handheld computer-based technologies to utilize pharmacopeias and clinical management guidelines, it is our belief that a risk calculator will find widespread use and assist physicians and surgeons in making clinical decisions.

Apart from identifying high-risk patients, we foresee the risk calculator as an important tool in the informed consent process. The process of patient-centered informed consent requires, in addition to patient capacity and freedom of action, the presentation of adequate information regarding risks and benefits generally defined using a reasonable person standard.¹¹ Accurate individualized assessment of MICA, which contributes greatly to morbidity and mortality, would certainly assist in meeting the latter objective. Physicians have long quoted the most current literature to explain risks of adverse outcomes associated with a procedure. This has not always been an easy task, because each patient is different, with a unique set of risk factors. Thus, this risk calculator will simplify the informed consent process by estimating the risk of MICA.

In spite of its many strengths, the present study has some limitations. Variables analyzed were limited to those recorded by NSQIP. Despite the data set being fairly comprehensive, with more than 50 preoperative variables, information on preoperative stress test, echocardiography, arrhythmia, and aortic valve disease was not available. Although aortic valve disease was a risk factor in the original risk index by Goldman et al,¹ it was not significant in the RCRI. Information on β -blocker use was not a part of the data set; however, this has not been significant in most previous indices.⁷ Outcomes for 1.3% of patients in the 2007 data set who stayed in the hospital beyond 30 days was not available. Unfortunately, known/remote CAD (except prior percutaneous coronary intervention and cardiac surgery) was not a variable in the data set, and this may have helped further increase the discriminative ability of the risk calculator.

There were some methodological differences in the present study compared with previous indices. The outcome studied in RCRI and Goldman's original risk index included pulmonary edema in addition to MICA, with the RCRI also including complete heart block; these were not outcomes in the present study because they are not a part of the NSQIP. The diagnosis of MI was slightly different in the present study than in previous similar studies. The current generation of troponin-based tests are much more sensitive than the creatinine kinase-based tests that were used in the RCRI and other previous risk indices. Hence, troponin-based tests are likely to identify any perioperative MI better than earlier tests. This may limit the predictive ability of previous risk indices that were not based on troponin testing, which is the current standard of care. Current guidelines define MI as elevation of troponin above the reference range. In the present study, a new elevation in troponin >3 times the upper level of the reference range in the setting of suspected myocardial ischemia was used as the definition. Although this would lead to the exclusion of some MIs, one can also argue that this makes MI a more definitive end point, because minor myocardial damage is excluded.

In conclusion, although fewer than 1% of patients develop MICA, 61% of these patients die within 30 days of surgery. This

high mortality rate seen in patients with MICA emphasizes the importance of risk stratification and preoperative optimization. This risk calculator with its high discriminative ability for intraoperative/postoperative MICA is a step in that direction.

Disclosures

None.

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CLINICAL PERSPECTIVE

Perioperative myocardial infarction or cardiac arrest (MICA) is associated with significant morbidity and mortality. The Revised Cardiac Risk Index (RCRI) is currently the most commonly used cardiac risk stratification tool; however, it has several limitations. It was derived and validated on a relatively small cohort of 4315 patients. Categorization of surgeries as used in the RCRI is not very relevant today because of advances in the field of surgery. Lastly, the RCRI lacks in its discriminative/predictive ability. The American College of Surgeons' National Surgical Quality Improvement Program is the largest and most complete prospective, clinical, national surgical database, with more than 200 hospitals participating. We analyzed this database to ascertain risk factors associated with intraoperative/postoperative MICA and to develop and validate a risk calculator for its prediction. Its performance was subsequently compared with that of the RCRI. Of the 211 410 patients, 1371 (0.65%) developed MICA. On multivariate logistic regression analysis, 5 predictors of MICA were identified: Type of surgery, dependent functional status, abnormal creatinine, American Society of Anesthesiologists' class, and age. The risk model based on the 2007 data set was subsequently validated on the 2008 data set (n=257 385) and used to develop an interactive risk calculator. The C statistic for our risk model was higher than that for the RCRI (0.884 versus 0.747), which implies a superior predictive ability. Furthermore, instead of classifying patients as high/intermediate/low risk for perioperative cardiac complications, we provided an exact model-based estimate of MICA probability for a patient. We anticipate the risk calculator will aid in surgical decision making and the informed consent process.